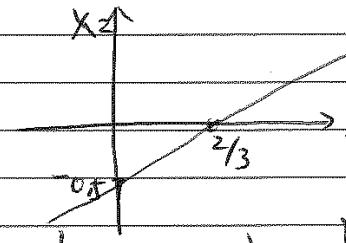
HW3

Q1

(a) $\vec{w}^T\vec{x} + w_0 = 0 \Rightarrow [-3 \;\; 4]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + 2 = 0 \Rightarrow -3x_1 + 4x_2 + 2 = 0$



$\hat{n} = \dfrac{\vec{w}}{\|\vec{w}\|}$, $\|\vec{w}\| = \sqrt{(-3)^2 + 4^2} = 5$, so $\hat{n} = \begin{bmatrix} -\frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$

(b) The signed distance from the point $\vec{x} = \vec{0}$ to the decision boundary is

$$r = \frac{y(\vec{x})}{\|\vec{w}\|} = \frac{\vec{w}^T\vec{0} + w_0}{\|\vec{w}\|} = \frac{w_0}{\|\vec{w}\|} = \frac{2}{5}$$

This is the signed distance from $\vec{0}$ to the boundary.
The signed distance from the decision boundary to $\vec{0}$ is $-\frac{2}{5}$.

(c) For $\vec{x}_A = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $y(\vec{x}_A) = [-3 \;\; 4]\begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 = -8$

$\vec{x}_A$ is on the $y < 0$ side (class C_2).

The signed distance from $\vec{x}_A$ to the decision boundary is

$$r_A = \frac{y(\vec{x}_A)}{\|\vec{w}\|} = -\frac{8}{5}$$

Q2

(a) $\sigma(a) = (1+e^{-a})^{-1}$, $\frac{d\sigma(a)}{da} = \frac{d(1+e^{-a})^{-1}}{da} = (-1)(1+e^{-a})^{-2}\frac{d(e^{-a})}{da}$

$= -\frac{-e^{-a}}{(1+e^{-a})^2} = \frac{1}{1+e^{-a}} \cdot \frac{e^{-a}}{1+e^{-a}} = \sigma(a)[1-\sigma(a)]$. //

(b) $E(\vec{w}) = -\sum_{n=1}^{N}\{t_n \ln y_n + (1-t_n)\ln(1-y_n)\}$

$\nabla_{\vec{w}} E(\vec{w}) = -\sum_{n=1}^{N} \nabla_{\vec{w}}\{t_n \ln y_n + (1-t_n)\ln(1-y_n)\}$

$= -\sum_{n=1}^{N}\{t_n \cdot \frac{1}{y_n}\nabla_{\vec{w}} y_n + (1-t_n)\frac{-1}{1-y_n}\nabla_{\vec{w}} y_n\}$

Since $y_n = \sigma(a_n)$, $\frac{dy_n}{da} = \sigma(a_n)[1-\sigma(a_n)] = y_n(1-y_n)$,

so $\nabla_{\vec{w}} y_n = \frac{dy_n}{da_n}\nabla_{\vec{w}} a_n = y_n(1-y_n)\nabla_{\vec{w}} a_n = y_n(1-y_n)\vec{\phi}(\vec{x_n})$.

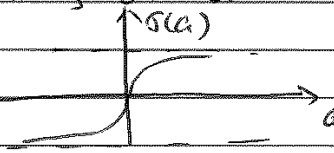
$\therefore \nabla_{\vec{w}} E(\vec{w}) = -\sum_{n=1}^{N}\{t_n \frac{1}{y_n} y_n(1-y_n)\vec{\phi}(\vec{x_n}) - (1-t_n)\frac{1}{1-y_n} \cdot y_n(1-y_n)\vec{\phi}(\vec{x_n})$

$= -\sum_{n=1}^{N}\{t_n(1-y_n)\vec{\phi}(\vec{x_n}) - (1-t_n)y_n\vec{\phi}(\vec{x_n})\}$

$= -\sum_{n=1}^{N}\{t_n\vec{\phi}(\vec{x_n}) - y_n\vec{\phi}(\vec{x_n})\}$

$= \sum_{n=1}^{N}(y_n - t_n)\vec{\phi}(\vec{x_n})$ //

(c) If the dataset is linearly separable, then there exists a $\vec{w}$ such that $\sigma(\vec{w}^T\vec{\phi_n}) > 0$ for all observations with $t_n = 1$ and $\sigma(\vec{w}^T\phi_n) < 0$ for all observations with $t_n = 0$.



Because of the shape of $\sigma(a)$:
$|\sigma(\vec{w}^T\phi_n)| < 1$ for finite $\vec{w}^T\phi_n$
This means that the prediction $y_n = \sigma(\vec{w}^T\phi_n)$ is not "perfect" — there is always "room for improvement" — by increasing $\|\vec{w}\|$, we can "push" y_n closer and closer to ± 1. As a result, the logistic sigmoid becomes a step function as $\|\vec{w}\| \to \infty$, i.e, the maximum likelihood $\vec{w}$ never converges to a finite weight vector.

Q3. For a loss matrix where $L_{kk} = 0$ and $L_{kj} = 1$ for all $k \neq j$,

(a) the expected loss is

$$\mathbb{E}[L] = \sum_{k=1}^{K} \sum_{j=1}^{K} \int_{R_j} L_{kj}\, p(\vec{x}, C_k)\, d\vec{x}$$

$$= \sum_{k=1}^{K} \sum_{\substack{j=1 \\ j \neq k}}^{K} \int_{R_j} 1 \cdot p(\vec{x}, C_k)\, d\vec{x} + \sum_{k=1}^{K} \int_{R_k} 0 \cdot p(\vec{x}, C_k)\, d\vec{x}$$

$$= \sum_{k=1}^{K} \sum_{\substack{j=1 \\ j \neq k}}^{K} \int_{R_j} p(\vec{x}, C_k)\, d\vec{x}$$

By definition, $\int_{R_j} p(\vec{x}, C_k)\, d\vec{x}$ for a given pair of k and j ($j \neq k$) is the misclassification probability of assigning $\vec{x}$ that is in C_k to C_j. The summation over all possible such pairs of j and k yields, by definition, the total misclassification (mistake) probability, i.e., $\mathbb{E}[L] = p(\text{mistake})$. //

(b) If a given $\vec{x}$ is assigned to C_1, the expected loss is

$$\mathbb{E}[L \mid \vec{x} \in R_1] = \underbrace{L_{11}}_{0}\, p(C_1 \mid \vec{x}) + L_{21}\, p(C_2 \mid \vec{x}) = L_{21}\, p(C_2 \mid \vec{x})$$

Similarly, $\mathbb{E}[L \mid \vec{x} \in R_2] = L_{12}\, p(C_1 \mid \vec{x}) + L_{22}\, p(C_2 \mid \vec{x}) = L_{12}\, p(C_1 \mid \vec{x})$.

That is, the expected loss for a single input $\vec{x}$, given our loss matrix, $L = \begin{bmatrix} 0 & L_{12} \\ L_{21} & 0 \end{bmatrix}$, solely comes from assigning it to C_1 when it should be C_2 ($\mathbb{E}[L \mid \vec{x} \in R_1] = L_{21}\, p(C_2 \mid \vec{x})$), or, assigning it to C_2 when it should be in C_1 ($\mathbb{E}[L \mid \vec{x} \in R_2]$). We make the optimal decision by picking the smaller loss — that is, we assign $\vec{x}$ to C_1 if $\mathbb{E}[L \mid \vec{x} \in R_1] < \mathbb{E}[L \mid \vec{x} \in R_2]$. This gives

$$L_{21}\, p(C_2 \mid \vec{x}) < L_{12}\, p(C_1 \mid \vec{x})$$

$$\Rightarrow L_{21}\,[1 - p(C_1 \mid \vec{x})] < L_{12}\, p(C_1 \mid \vec{x})$$

$$\Rightarrow p(C_1 \mid \vec{x}) > \frac{L_{21}}{L_{12} + L_{21}}, \text{ or, } \theta > \frac{L_{21}}{L_{12} + L_{21}}$$

When $L_{12} = L_{21}$, this is just $\theta > \frac{1}{1+1}$, or, $\theta > 0.5$. //

HW

Q4 Confusion matrices for model A, and model B:

(a)

Model A	Predicted Healthy	Predicted Diseased
True ~~Neg~~ Healthy	$N_{TN} = 99,900$	$N_{FP} = 0$
True Diseased	$N_{FN} = 100$	$N_{TP} = 0$

model B	Predicted Healthy	Predicted Diseased
True Healthy	$N_{TN} = 99,580$	$N_{FP} = 320$
True Diseased	$N_{FN} = 20$	$N_{TP} = 80$

Accuracy of model A $= \frac{N_{TP}+N_{TN}}{N} = \frac{99,900}{100,000} = 99.9\%$

Accuracy of model B $= \frac{N_{TP}+N_{TN}}{N} = \frac{80+99,580}{100,000} = 99.66\%$

Judging from accuracy alone, model A is superior.

b) Precision of Model A $= \frac{N_{TP}}{N_{TP}+N_{FP}} = 0$

- - - - B $= \frac{80}{80+320} = 20\%$

Recall (sensitivity) of Model A $= \frac{N_{TP}}{N_{TP}+N_{FN}} = 0$

- - - - B $= \frac{80}{80+20} = 80\%$

F-score of model A $= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 0$

- - - - B $= \frac{2 \times 0.2 \times 0.8}{0.2+0.8} = 32\%$.

c) Since Accuracy $= \frac{N_{TP}+N_{TN}}{N_{TP}+N_{TN}+N_{FP}+N_{FN}} = \frac{N_{TP}/N_{TN}+1}{N_{TP}/N_{TN}+1+N_{FP}/N_{TN}+N_{FN}/N_{TN}}$

When $N_{TN} \gg N_{TP}, N_{FP}, N_{FN}$, all terms (except the "1") in the numerator and the denominator approach 0, resulting in high accuracy despite the model's failure to detect positive cases. The F-score explicitly excludes N_{TN}, preventing inflation by TN cases. The F-score is high only if both precision and recall are high.